

AS/400 → Snowflake Operating Model

Enterprise Data Platform Operating Framework

1. Operating Model Overview

Purpose: Define how the Snowflake platform will operate post-modernization, replacing AS/400 operational patterns with cloud-native processes.

Core Pillars:

- Platform Ownership
- Data Product Ownership
- Ingestion & Transformation Operations
- Governance & Security
- FinOps & Cost Management
- Support & Incident Management
- Continuous Improvement

Guided Links:

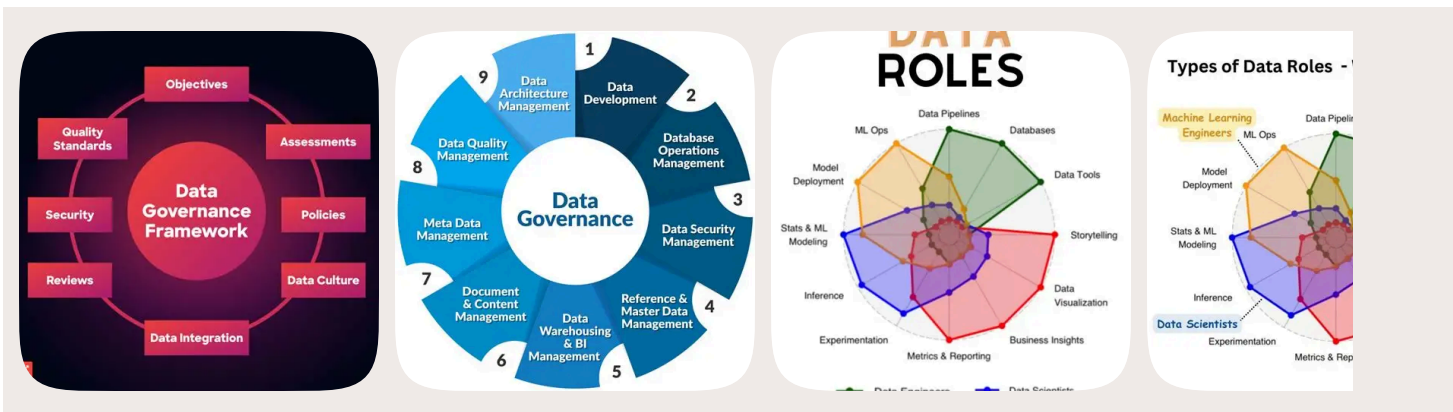
- [Modern data platform architecture](#)

2. Platform Ownership Model

Objective: Establish clear ownership for Snowflake as a strategic enterprise platform.

Roles & Responsibilities

- **Enterprise Architecture (EA)** — Owns platform standards, patterns, and roadmap
- **Cloud Ops** — Owns infrastructure, networking, storage, and Snowflake account operations
- **Data Engineering** — Owns ingestion pipelines, CDC, and ELT frameworks
- **DBT Lead** — Owns transformation logic, lineage, CI/CD
- **Security** — Owns RBAC/ABAC, masking, compliance
- **Business Data Owners** — Own data quality, definitions, and acceptance



Guided Links:

- [Snowflake RBAC best practices](#)

3. Ingestion Operating Model

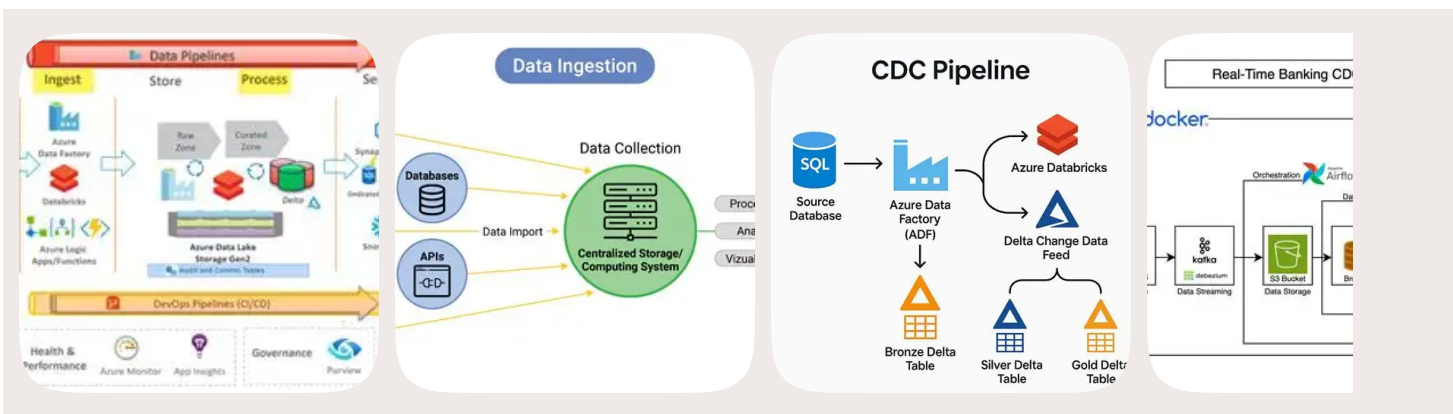
Objective: Define how data flows into Snowflake continuously and reliably.

Ingestion Modes

- Full Loads: Scheduled batch ingestion
- CDC Loads: Journaling → Streams & Tasks
- Real-Time: Snowpipe auto-ingest

Operational Responsibilities

- Data Engineering: Build & maintain ingestion pipelines
- Cloud Ops: Monitor landing zone health
- Security: Validate encryption & access policies
- QA: Validate ingestion accuracy



Guided Links:

- [Snowpipe ingestion](#)
- [CDC journaling](#)

4. Transformation Operating Model (DBT-Driven)

Objective: Replace RPG/CL logic with transparent, modular SQL transformations.

DBT Responsibilities

- **Staging Models:** Type casting, renaming, cleaning
- **Intermediate Models:** Business logic, joins
- **Marts:** Facts, dimensions, SCD Type 1/2
- **Testing:** Schema, uniqueness, referential integrity
- **CI/CD:** Automated deployments

Execution Cadence

- Scheduled DBT runs
- Event-driven runs (CDC)
- On-demand runs for hotfixes

Guided Links:

- [DBT transformation models](#)

5. Governance & Security Operating Model

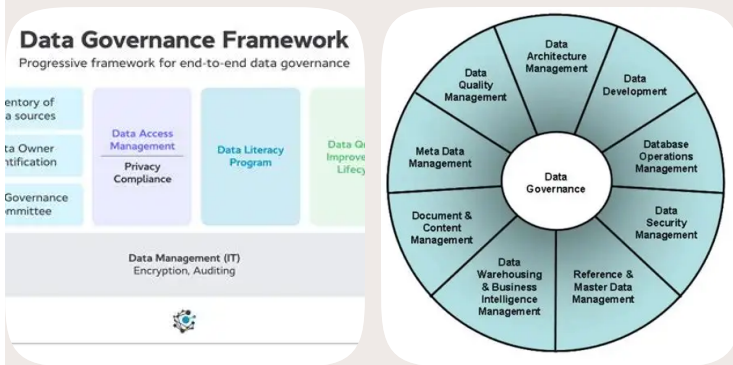
Objective: Ensure compliance, auditability, and secure access across the platform.

Governance Components

- Data catalog & lineage
- Data quality SLAs
- Data ownership model
- Metadata management
- Audit logging

Security Components

- RBAC/ABAC
- Masking policies
- Row-level security
- Object tagging
- Access reviews



Guided Links:

- [Data classification](#)

6. FinOps & Cost Management Operating Model

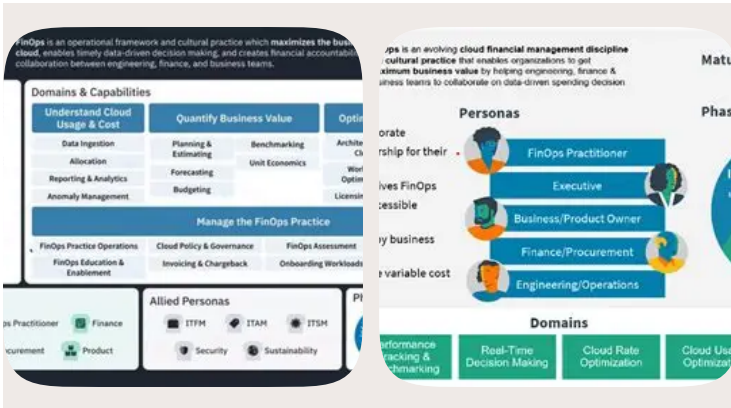
Objective: Control Snowflake spend while maximizing performance.

Cost Controls

- Warehouse right-sizing
- Auto-suspend/auto-resume
- Resource monitors
- Query optimization
- Storage tiering

FinOps Cadence

- Weekly cost dashboards
- Monthly optimization reviews
- Quarterly architecture tuning



Guided Links:

- [Snowflake cost optimization](#)

7. Support & Incident Management

Objective: Provide reliable, predictable operational support.

Support Tiers

- L1: Monitoring, ingestion failures, Snowpipe alerts
- L2: DBT job failures, CDC issues, schema drift
- L3: Architecture issues, performance tuning, security escalations

Incident Workflow

- Detection → Triage → Root Cause → Fix → Preventive Action

Guided Links:

- [Data validation](#)

8. Continuous Improvement Operating Model

Objective: Evolve the platform through iterative enhancements.

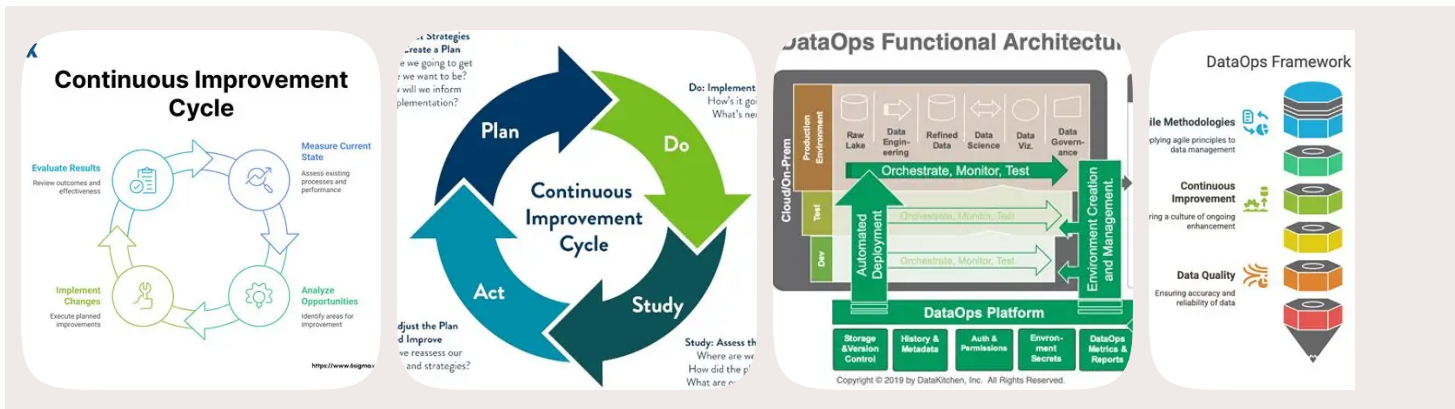
Improvement Streams

- Performance tuning
- New data products

- New ingestion pipelines
- Governance enhancements
- Automation & observability upgrades

Cadence

- Monthly DataOps review
- Quarterly platform roadmap updates
- Annual architecture refresh



9. Operating Model RACI Summary

Objective: Clarify ownership across all operational domains.

Domain	EA	Cloud Ops	Data Eng	DBT Lead	Security	QA	Business
Platform Standards	R	C	C	C	C	I	I
Ingestion Pipelines	C	C	R	I	I	C	I
Transformations	C	I	C	R	I	C	C
Governance	R	I	C	C	C	C	A
Security	C	C	I	I	R	I	I
Cost Management	C	R	C	C	C	I	I

Domain	EA	Cloud Ops	Data Eng	DBT Lead	Security	QA	Business
Data Quality	C	I	C	C	I	R	A

Guided Links:

- [AS/400 → Snowflake RACI Matrix](#)

10. Final CIO Summary

The AS/400 → Snowflake Operating Model delivers:

- A **cloud-native, governed, scalable** data platform
- Clear ownership across engineering, governance, and business
- Automated ingestion, transformation, and validation
- Strong security and compliance posture
- Predictable cost management
- A foundation for AI/ML and enterprise analytics